COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure & Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	Jul 29, 2026	**DATE SUBMITTED:**	Jul 29, 2026
NAME:	Bal, Kevin Ranz J.	**INSTRUCTOR:**	Engr Jimlord Quejado

Seatwork 4.1 **Array-Based vs. STL Stack Implementations**

1. PROCEDURE OUTPUT

What procedures did you perform?

PROCEDURE:

Procedure that performed:

Performed answering the new topic about stacks and answer ILO A & ILO B

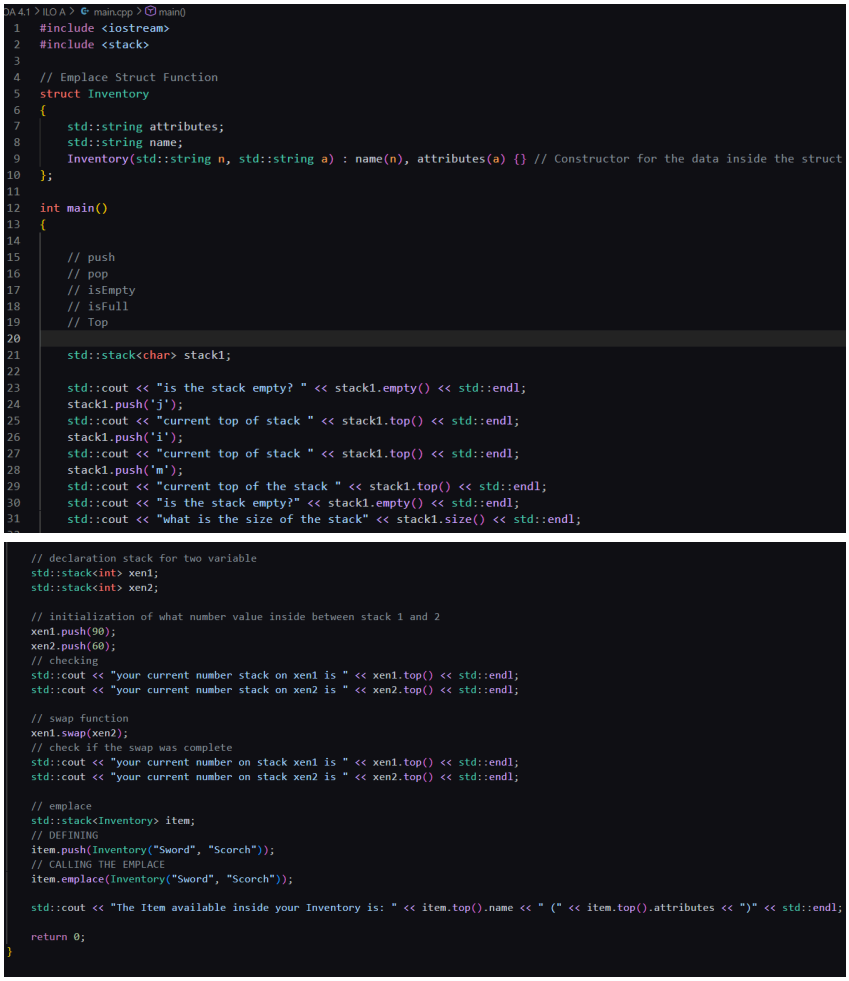
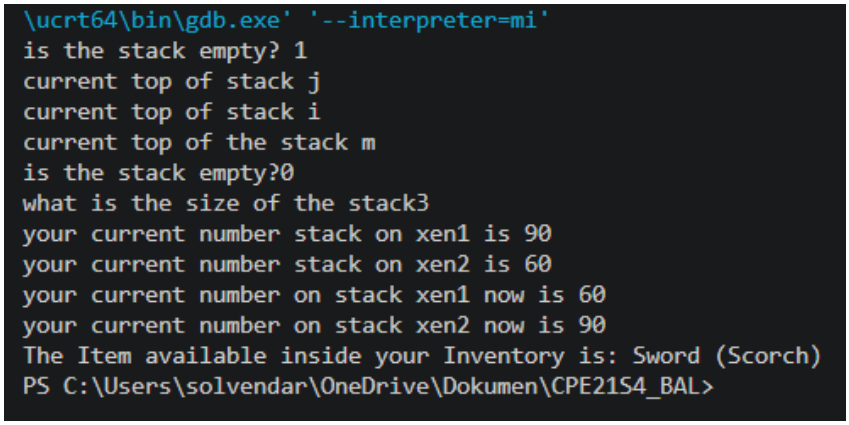
ILO A

Task Title & Task Objective

- **Implement a stack using the C++ STL (`std::stack`).**

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Screenshot of the code	
Output of the code	

```
\ucrt64\bin\gdb.exe' '--interpreter=mi'
is the stack empty? 1
current top of stack j
current top of stack i
current top of the stack m
is the stack empty?0
what is the size of the stack3
your current number stack on xen1 is 90
your current number stack on xen2 is 60
your current number on stack xen1 now is 60
your current number on stack xen2 now is 90
The Item available inside your Inventory is: Sword (Scorch)
PS C:\Users\solvendar\OneDrive\Dokumen\CPE21S4_BAL>
```

Analysis	In this code, it teaches us how stack works. In this figure, we include a header first “<stack” so the function can be used. And next, we created a stack that inside was a char with a variable name stack1 and used how each function works. On the push () function, it inserts a value that depends on the declaration and to the user while the pop() removes the item from the stack. IsEmpty and isFull only observe if there's data inside on the stack that returns true if stack is empty and else if false, while top() was displaying or returns the top element of the stack. In this ILO A, we created a char stack and it checks first if the stack is empty, after that, it checks the current top of the stack three times and after each call it push() a letter “j i m” and lastly, it checks the size of the stack and give an output of 0. In swap function, I created two stack as an int and called xen1 and xen2 and after that, I pushed a value inside both of them and also create a cout where it checks their current number and i swapped the value of xen1 to xen2 using xen1.swap(xen2) and checks their value again by calling their function xen1.top and xen2.top. On emplace function, I created a struct where there's a data inside as 2 string that I’ll use in methods in main drive and I created a constructor function where there’s a parameter that calls the two data inside the struct and in main driver, I declared a stack and inside of it was the Inventory which is the constructor / function to call the data inside the struct and use the struct variable to push and inside was the inventory function and including the data i want to input inside and use the emplace function to successfully implement it inside the stack variable once I called it after and creating a top function but i also called the variable name inside the struct because the vscode misbehave and cannot read my emplace code function once I run it.

ILO B

Task Title & Task Objective
Implement a stack using a Static Array.

Screenshot of the code

HOA 4.1 > ILO B > main2.cpp > DisplayAll()

```cpp
#include <iostream>
#include <stack>

// DEFINITION CHOICE
// #define maxSize 10 (method 1)
// const int maxSize = 10; (method 2)
// const maxSize_t = 10;

// global variable
#define maxSize 10
int stackArr[maxSize];
int top = -1, newData;

// function prototype
// Top, Push, isEmpty, isFull

// observer
bool isEmpty();
bool isFull();
void Top(); // T always capital
void DisplayAll();

// transformer operators
void push();
void pop();

int main() {

    while (true) {

        int choice;

        std::cout << "===========================================\n";
        std::cout << "                Stack Operations: " << std::endl;
        std::cout << "[1] push" << std::endl;
        std::cout << "[2] pop" << std::endl;
        std::cout << "[3] isEmpty" << std::endl;
        std::cout << "[4] isFull" << std::endl;
        std::cout << "[5] Top" << std::endl;
        std::cout << "[6] Display All" << std::endl;
        std::cout << std::endl;
        std::cout << "Input your choice ";
        std::cin >> choice;
```

```
44
45          switch (choice) {
46
47          case 1:
48              push();
49              break;
50          case 2:
51              pop();
52              break;
53          case 3:
54              isEmpty();
55              break;
56          case 4:
57              isFull();
58              break;
59          case 5:
60              Top();
61              break;
62          case 6:
63              DisplayAll();
64              break;
65          default:
66              std::cout << "Invalid choice." << std::endl;
67          }
68      }
69      return 0;
70  }
71  // function definition
72  bool isEmpty(){
73      return (top == -1);
74  }
75
76  bool isFull(){
77      return (top == maxSize - 1);
78  }
79
80  void Top() {
81
82      // prints the top element
83      std::cout << "The top element is: " << stackArr[top] << std::endl;
84  }
85
```

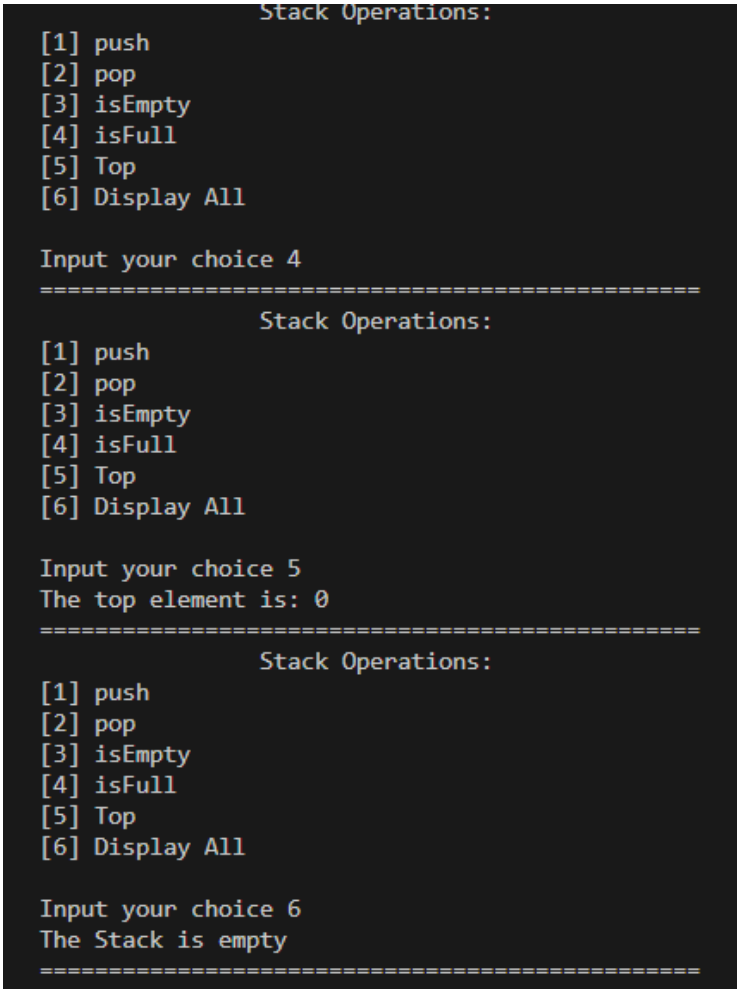
Output of the code

```
==================================================
                Stack Operations:
[1] push
[2] pop
[3] isEmpty
[4] isFull
[5] Top
[6] Display All

Input your choice 1
==================================================
                Stack Operations:
[1] push
[2] pop
[3] isEmpty
[4] isFull
[5] Top
[6] Display All

Input your choice 2
Stack Underflow!
==================================================
                Stack Operations:
[1] push
[2] pop
[3] isEmpty
[4] isFull
[5] Top
[6] Display All

Input your choice 3
==================================================
                Stack Operations:
```



Analysis	In this code, it defines a constant maxSize of an array that cannot be changed if ever gets touched and creates a top value where inside was negative 1 because the stacks start reversely and the array starts from zero. It creates a four function which isEmpty which inside was only the method that returns the -1 once it equals to the top variable that declared and on the isFull, it returns the maxSIze -1 if it got equals to the top variable, and on the top function it prints the top element of the stack of the stackArr. While on the push method, it creates an error catching to check also the stack if it's full or empty and use an if-else statement to check the isFull function to verify if its full, it will display “stack overflow”, while the other function of it was a user input where it let the user input its want to push or add and the stackArr will increment using the top variable above it where inside was the newData. On pop function, it creates a error catching where it checks if its empty using if statement and display “stack if empty” if the condition will be true and checks it if empty and the stackArr where inside of it was the top variable that the user inputted and remove the element from the top variable using decrement. In last function which it will display it all, it

	checks first using if statement and inside was the isEmpty function and for displaying, it creates a for loop where i is equals to top and top should be greater or equal to it and it will decrement to reads and prints the element at the current position. In driver code, it created a while loop so the loop will never stop as long as the choice will equal to false. Inside it was a variable choice for the user input and creates a switch case, and each cases was the function / method that created outside the driver that will run once the user choice will equal to it and breaks if out of 6 choices was not of the users choice but the program will still run because the choice doesn't declared as false.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

SUPPLEMENTARY:

ILO C

Task Title & Objective:

Screenshot of the code	
Output of the code	
Analysis	

Conclusion

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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